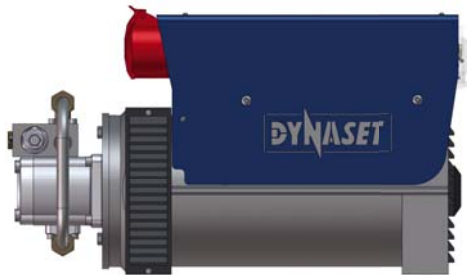
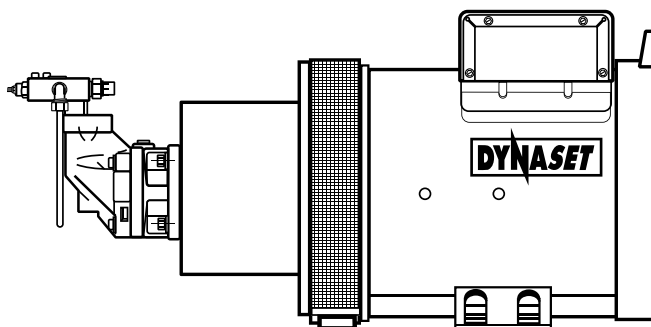
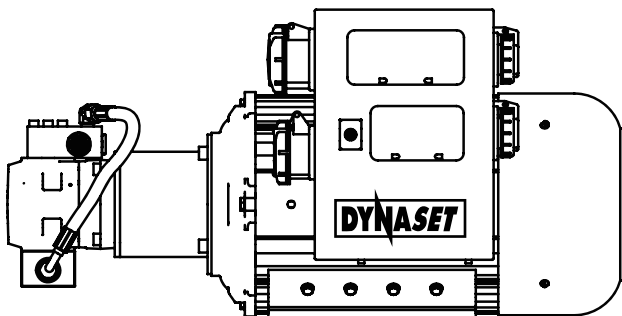


# HG

## HYDRAULIC GENERATORS



## OPERATING INSTRUCTIONS



|            |  |
|------------|--|
| HG 1,0 kVA |  |
| HG 2,0 kVA |  |
| HG 3,5 kVA |  |
| HG 5,0 kVA |  |
| HG 6,5 kVA |  |
| HG 10 kVA  |  |
| HG 12 kVA  |  |
| HG 15 kVA  |  |
| HG 20 kVA  |  |
| HG 30 kVA  |  |
| HG 40 kVA  |  |
| HG 50 kVA  |  |
| HG 60 kVA  |  |
| HG 70 kVA  |  |

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**SAFETY PRECAUTIONS**

Operating voltage of HG-unit varies 110 - 415V depending on generator type.

The hydraulic system is usually pressurised up to 210 (420) bar.

In order to avoid damages and accidents operator and maintenance personal must act in compliance with the laws, regulations and recommendations issued by local electricity and work safety authorities.

Follow all your local safety instructions related to electric devices and the high pressure hydraulics.

**TECHNICAL CONDITION OF YOUR MACHINERY AND EQUIPMENT MUST BE SUBJECTED TO CONSTANT SURVEILLANCE.**

Hydraulic system of a carrier machine should be maintained according to the service program.

All couplings, valves and hoses should be maintained tight and clean as well as kept under constant observation.

Leaks in the hydraulic system must be fixed immediately to avoid injuries caused by high pressure and oil blowouts.

Prior to maintenance, detaching from a carrier or disassembling a DYNASET-unit, the hydraulic system of a carrier machine should be stopped and DYNASET's hydraulic circuit depressurized.

When operating or maintaining in the immediate vicinity of DYNASET hydraulic generator unit or accessories, appropriate protective clothing, safety goggles, gloves, ear and eye protection should be worn. Do not touch parts heated by hydraulic oil.

**EXTERME CLEANLINESS MUST BE MAINTAINED WHEN CARRYING OUT ANY DISASSEMBLING OR REPAIR OF HG UNIT AND HYDRAULIC SYSTEM. THIS IS CRUCIAL TO ENSURE SAFE, RELIABLE AND LONG-LIFE OPERATION OF YOUR EQUIPMENT.**

All installation and service of both hydraulic and electric equipment must be performing experienced personnel only.

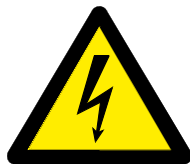
**WARNING STICKERS**

Dynaset dispatch department includes one (1) warning sticker bag per one (1) main product.

Product recipient is obligated to fix determinate warning sticker to Dynaset product.

Attach sticker to visible and appropriate place or close to Dynaset product where it's easily seen.

Before attaching sticker clean surface with solvent detergent.



READ OPERATING INSTRUCTIONS. RISK OF ELECTRIC SHOCK. BEWARE OF HOT SURFACE. USE EAR PROTECTOR.

## GENERAL NOTES

DYNASET hydraulic generators are compact and integrated all-in-one units, especially designed for mobile installation. The only power source is a hydraulic system to provide generator with required hydraulic fluid flow at demanded pressure.

DYNASET hydraulic generator transforms hydraulic power into a high quality electricity, which is to be used for energising electrical tools, appliances and other equipment.

## CONSTRUCTION

DYNASET hydraulic generator is comprised of hydraulic motor and alternator. The power-to-weight ratio of DYNASET hydraulic generator is excellent due to modern mono-bearing construction when the other end of rotor's shaft is supported by hydraulic motor. (Generators over 20 kVA are bi-bearing units). Unit's rotor, connected to the hydraulic motor's shaft, is provided with an excitation windings. Electric power is taken from stator windings. Auxiliary winding of voltage control system is located in generator's stator as well. Windings are isolated from unit's body and their insulation class is H. The unit has an automatic frequency control valve with ports for pressure and return lines.

Protection class of standard DYNASET hydraulic generators complies with specifications IP 23 or IP 34. Units with protection class IP 54 are available by request.

Electric sockets are standard or special when requested. Automatic circuit breakers protect unit from overload. DYNASET hydraulic generators are self-excited. The electric load can be connected prior to starting a generator. DYNASET hydraulic generator is assembled in lightweight aluminium alloy molded case with footing for fixing.

## AUTOMATIC VOLTAGE CONTROL

Automatic voltage regulator adjusts rotor's excitation power, maintaining constant output voltage at discontinuous electric load.

Voltage control function depends on the rotor's rotation speed, so that the nominal voltage value is being set at nominal frequency (except electronic voltage regulator).

### SINGLE PHASE GENERATORS:

Voltage control of single phase generator is effected with a capacitor circuit, connected to an auxiliary winding. Voltage control keeps the voltage constant through the whole load range with accuracy of  $\pm 6\%$ . Performance speed of voltage control is less than 0,1 sec.

### 3-PHASE GENERATORS:

3-phase generators are equipped either with compound or electronic voltage regulator.

Compound regulator is connected to the auxiliary winding and maintains the output voltage constant through the entire load range with accuracy of  $\pm 5\%$ . Compound regulator sets the excitation current according to electric load of each phase individually with its separate current windings. Each phase can be loaded up to maximum current deliberately.

Electronic voltage regulator is connected to the stator's windings and achieves an accuracy of  $\pm 3\%$ .

Electronic voltage regulator constantly compares output voltage to the pre-adjusted reference value and sets the excitation current according to the load.

Performance speed of both compound and electronic voltage controls is less than 0,1 sec.

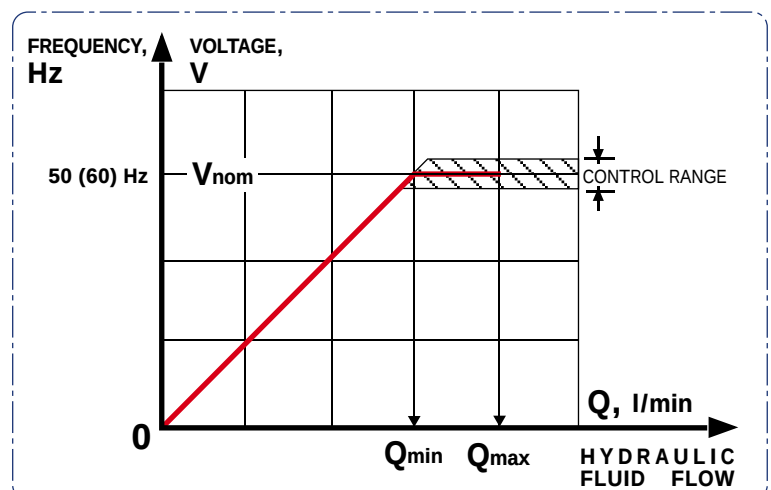
## PERFORMANCE OF AN AUTOMATIC FREQUENCY CONTROL VALVE

Automatic frequency control valve is to keep both rotation speed and frequency constant ( $\pm 5\%$ ) when incoming hydraulic oil flow (Q) can vary from minimal (= nominal demanded flow  $Q_{nom} + 1 \dots 5$  l/min) up to value exceeding  $Q_{min}$  by 10 – 30 l/min depending on generator size.

$$Q_{min} = Q_{nom} + 1 \dots 5 \text{ l/min}$$

$Q_{min}$  - MINIMAL HYDRAULIC FLOW

$Q_{nom}$  - NOMINAL HYDRAULIC FLOW



## CLASSIFICATION

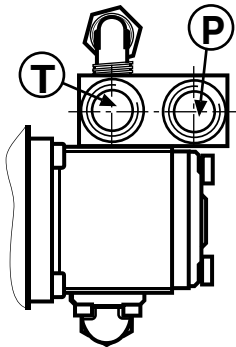
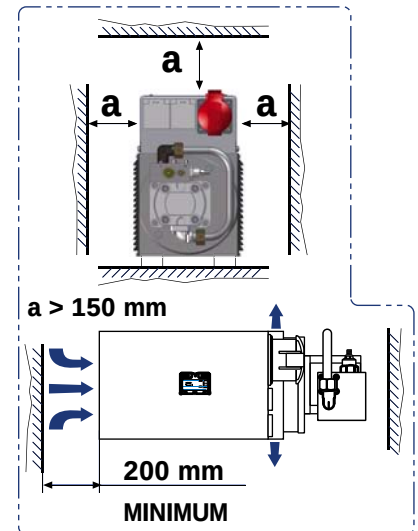
DYNASET hydraulic generators are manufactured in conformity with the 98/37, 73/23, 89/336 CEE directives and their amendments. They are also manufactured in compliance with the following regulations: CEI 2-3, EN 60034-1, IEC 34-1, VDE 0530, BS4999-5000, CAN/CSA-C22.2, NF 51.100 and N°14-95 - N°100-95. By request DYNASET hydraulic generators can be equipped with a radio interference protection to meet requirements of specifications MIL STD 461 A/B and VDE 0875 class N.

## INSTALLATION

DYNASET hydraulic generator works installed into an original hydraulic system of any carrier machine and can be placed deliberately ensuring sufficient venting and an easy access to the unit. Position of a generator should be horizontal.

Additionally, return line must be connected to a hydraulic oil tank directly in *the shortest possible way* in order to keep return oil pressure under 5 bar. Generally DYNASET's T-line is to be connected directly to the return line of a hydraulic system, when hydraulic fluid flows to the tank only through the return filter and cooler. Both filtering degree and cooling capacity of a hydraulic system should be sufficient (ref. to PARAMETERS).

It is possible to use DYNASET either as a mobile generator and connect hydraulic power with quick couplings, or make a fixed installation on a carrier machine. With DYNASET generator installed on your vehicle all electric power tools come easily along with you to any place.



To put DYNASET generator into operation only pressure- (P) and return (T) lines of a hydraulic system are to be connected to the unit's hydraulic ports. Ensure that the oil flow is sufficient to run the unit, i.e. at least the minimal flow must be available.

At the oil flow less than demanded nominal rate the unit will not work properly at all. If the oil flow is so excessive, that the automatic speed control valve is unable to handle it, the unit starts hunting. In the latter case the oil flow is to be reduced either by dropping down the rotation speed of hydraulic pump or using flow limiter valve. **DYNASET priority valve is recommended.**

After having been connected to the hydraulic system, DYNASET hydraulic generator is to be started by directing the oil flow to the unit's pressure port by means of control valve - manual or solenoid. Even simple ball valve or three-way valve can be used.

To ensure DYNASET's proper operation, the hydraulic pressure must be set to the level, at which at least the nominal power output is achieved (ref. to PARAMETERS). Lower pressure reduces output power. Maximum hydraulic pressure is 210 bar (ref. to PARAMETERS). If the hydraulic system main pressure is higher, service pressure relief valve must be used.

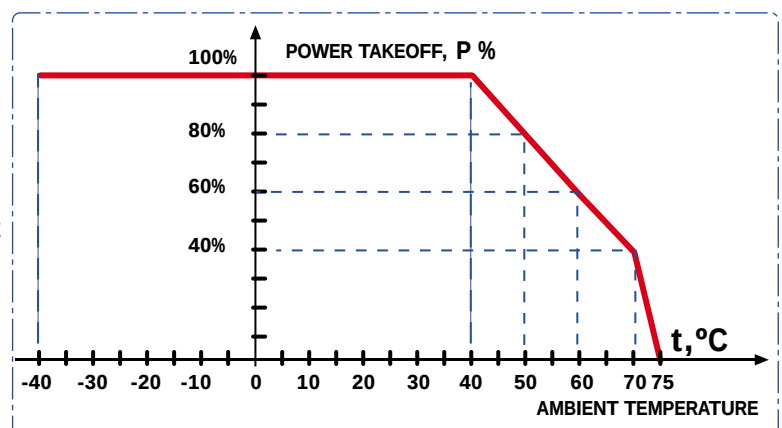
Ensure also that your machine oil cooling is working properly when DYNASET is in operation.

DYNASET hydraulic generators are designed for easy integration into hydraulic system of any type.

## AMBIENT TEMPERATURE

To avoid the power loss, it is not recommended to use DYNASET generator when the ambient temperature exceeds +40 °C. At the ambient temperature exceeding +40 °C power takeoff should be limited in accordance with the attached diagram (for instance, at the ambient temperature of +50 °C the power takeoff should not be more than 80 % of the maximum).

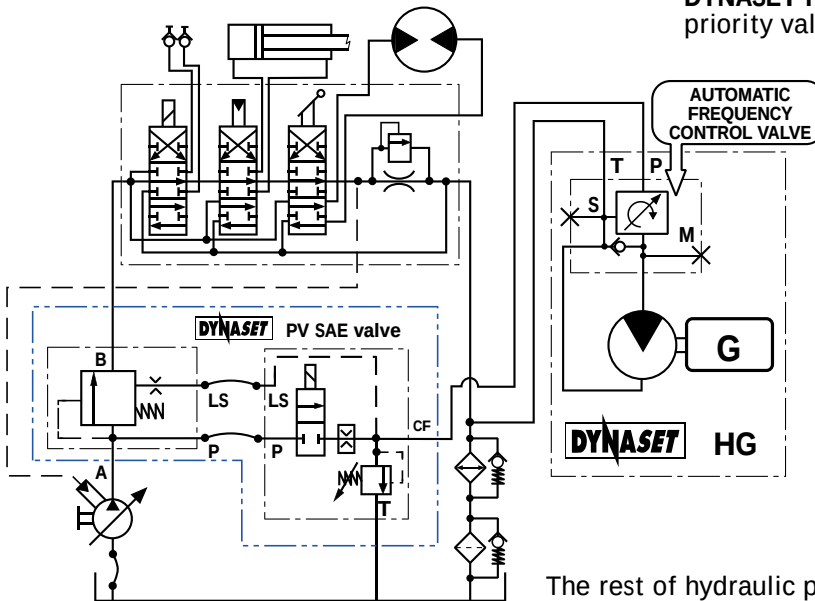
If the ambient temperature achieves +40 °C, mechanical ventilation can be arranged to ensure full power output of the generator.



## ... INSTALLATION

### 1. OPEN CENTRE HYDRAULIC SYSTEM WITH VARIABLE DISPLACEMENT PUMP

The demanded hydraulic flow is to be ensured and controlled with a DYNASET PRESSURE COMPENSATED PRIORITY VALVE.



**DYNASET PV SAE 3/4 - 1 1/4 -XX lpm - 12/24 V** priority valve includes following components:

1. Sandwich-mounted pressure compensator with SAE-flange specification;
2. Solenoid valve 12/24V;
3. Flow limiter;
4. Pressure relief valve.

PV SAE priority valve is designed for the installation to main pressure line between SAE-flanges of main hydraulic pump. Pre-adjusted, independent from other functions and prioritised hydraulic flow for DYNASET-unit comes from the solenoid valve.

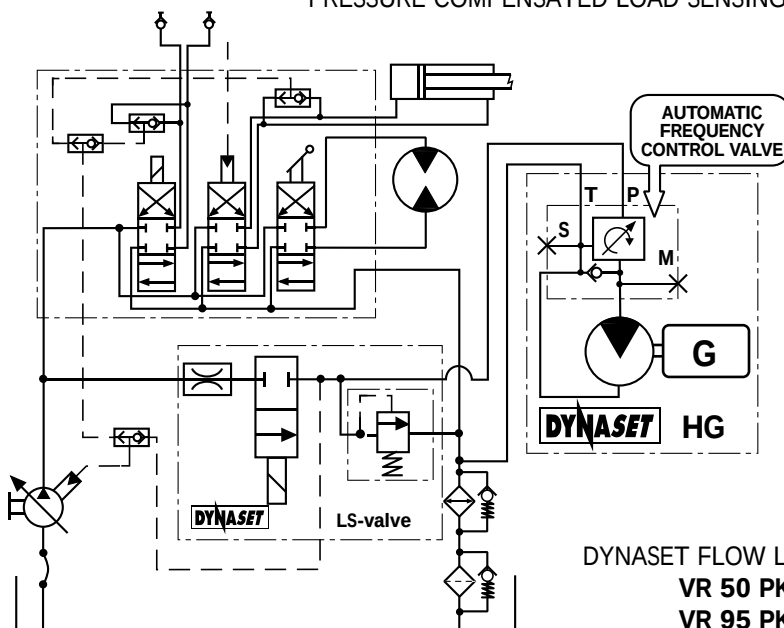
The rest of hydraulic pump's capacity is available for all other functions. Furthermore, pump's control works together with DYNASET PV SAE priority valve.

ALSO AVAILABLE for open centre hydraulic systems with variable displacement pump: PRIORITY VALVE PV C3C 3/4 - 1 1/4 - 12/24 V.

DYNASET PRIORITY VALVE enables to operate your DYNASET-unit simultaneously with other hydraulic executors.

### 2. CLOSED CENTRE HYDRAULIC SYSTEM WITH VARIABLE DISPLACEMENT PUMP

The demanded hydraulic flow is to be ensured and controlled with a DYNASET PRESSURE COMPENSATED LOAD SENSING (LS) VALVE.



DYNASET LS-valve includes following components:

1. Flow limiter;
2. Solenoid valve 12/24V;
3. Pressure relief valve;
4. Shuttle valve (OPTIONAL).

DYNASET LOAD SENSING VALVES:

**LSV 40, LSV 60, LSV 95 and LSV 150**

LS-connection, pressure relief, max. hydraulic flow 40, 60, 95 ja 150 l/min respectively.

DYNASET FLOW LIMITERS to the pressure line:

**VR 50 PK - 1/2**, max. 50 l/min, with pressure compensation;

**VR 95 PK - 3/4**, max. 95 l/min, with pressure compensation;

**VRD 180 PK - 1**, max. 180 l/min, with pressure compensation.

ALSO AVAILABLE for closed centre hydraulic systems with variable displacement pump:

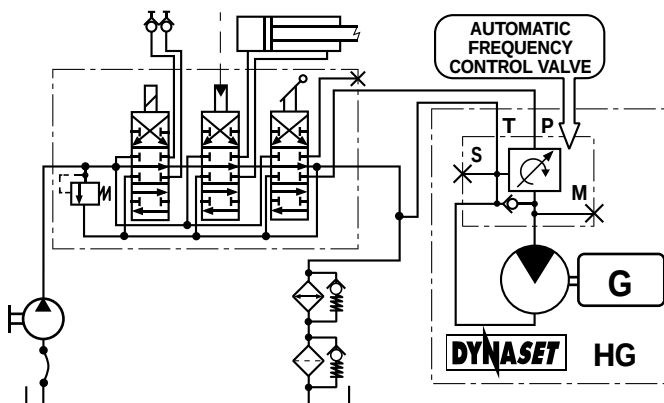
DYNASET SOLENOID VALVE to the pressure line for remote starting.

**SV 70 NC - 1/2 - 12 / 24 V** max 70 l/min with LS -connection;

**SV 150 NC - 1 - 12 / 24 V** max 150 l/min with LS -connection.

## ... INSTALLATION

### 3. HYDRAULIC SYSTEM WITH CONSTANT DISPLACEMENT PUMP



The demanded oil flow is to be ensured with proper hydraulic pump choice. In systems with redundant hydraulic flow an installation with **DYNASET PRIORITY VALVE** is recommended.

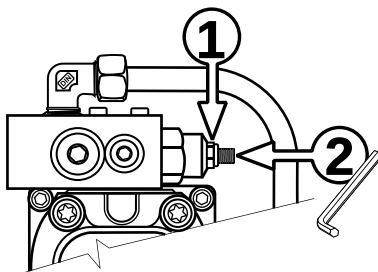
Installation with a standard pressure compensated 3-way valve should be avoided in order to close off potential waving in hydraulic system.

ALSO AVAILABLE for hydraulic systems with constant displacement pump:  
DYNASET FREE CIRCULATION VALVE with solenoid and pressure control, type **VKV 90–3/4**.

### OUTPUT FREQUENCY ADJUSTING AT START-UP

DYNASET hydraulic generator is tested and adjusted at factory but it is recommended to check the output frequency prior to taking the unit into operation.

- 1**  
Ensure that the generator is properly connected to the hydraulic system of your carrier machine and there are no oil leakages in system.
- 2**  
Start the engine of your carrier machine. Adjust the engine speed to the demanded level if necessary.
- 3**  
Start the generator with a control valve.
- 4**  
Check output frequency and tune it if necessary. For this purpose universal multimeter can be used. When the generator is running without load, switch the metering device to the Hz-position and check the frequency from each socket.



- 5**  
If the frequency is out of allowed range:
  - 5.1  
ensure whether the hydraulic fluid flow is on the demanded level.
  - 5.2  
set the frequency by adjusting the RPM-cartridge:

Loosen the checknut (1), make an adjustment (2) with adjusting screw according to the readings of frequency meter and then lock the checknut.

If a multimeter with frequency measurement is not available, operation of the generator can be tested by measuring the output voltage.

#### NOTE !

When making adjustment, the hydraulic fluid should be at normal operation temperature !

#### IF DYNASET GENERATOR IS INSTALLED TO HYDRAULIC SYSTEM WITHOUT DYNASET INSTALLATION VALVE:

Usually installation to the hydraulic system is being made with traditional pressure compensated flow control valve.

**Regardless of hydraulic system—whether is of open or closed centre type - the setting should be made in following way to ensure proper installation of your generator:**

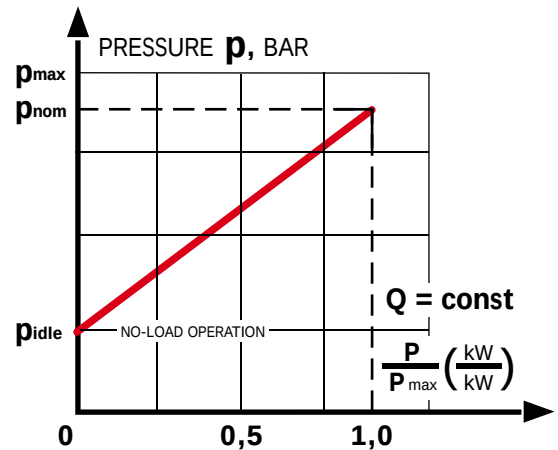
1.  
Start your machine and switch on the DYNASET generator.
2.  
Release the checknut 1 and turn the adjusting screw 2 clockwise for a couple of revolutions.
3.  
Adjust the hydraulic fluid flow with pressure compensated flow control valve until the output frequency will achieve the value of 51–52 (61–62 Hz). Follow the readings of multimeter.
4.  
Finalise the setting: turn the adjusting screw 2 counter-clockwise until it will take the flow under control—you'll feel it !  
Then turn the screw back for a quarter of revolution and lock the setting with checknut 1.

## OPERATION

After ensuring the proper operation of the generator, the load can be connected. From now on the load can be connected prior to starting the unit due to self-excitation characteristics of DYNASET generator.

The load can be applied by switching on any power consumer, such as work lamp, grinder, electric welder, electric motor etc., plugged into generator sockets. The pressure in hydraulic system is being adjusted according to the power output demand, when the hydraulic fluid flow is being kept constant.

DYNASET hydraulic generator produces high quality electric power within pressure range from **idle run pressure** to the maximum allowed value, usually 210 or 420 bar. The nominal power output is achieved at pressure value, which is noticeably less than the maximum allowed operation pressure. Refer to the diagram above and technical specifications.



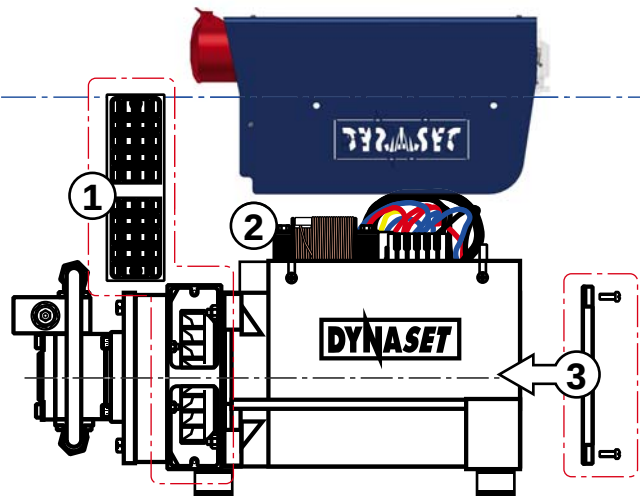
## MAINTENANCE

DYNASET hydraulic generators are low-maintenance units. Only normally wearing parts such as sealings in hydraulics, brush collector and bearings should be replaced when necessary. Refer also to the TROUBLESHOOTING section.

**ATTN. !**  
**CLEANLINESS OF YOUR HG-UNIT MUST BE MAINTAINED ON A REGULAR BASIS. CHECK YOUR EQUIPMENT AFTER EVERY WORKING SHIFT AND DEPENDING ON AN OPERATIONAL ENVIRONMENT CLEAN A GENERATOR AS FREQUENTLY AS NECESSARY TO KEEP IT IN PERFECT WORKING CONDITION.**



Use blow gun with a **REASONABLE DELIBERATION** to dust your equipment.



Remove side screens (1) and air diffuser (3) to clean fan compartment as well as rotor and stator. Remove cover (2) to dust alternator's electric components. Ensure electric enclosure's drain holes are dirty free. Having dusted/cleaned the generator, replace screens/covers and secure them with appropriate screws. (Above picture for reference only - refer to the data pages of specific HG-model).

Remove all oil deposits. Remove all unnecessary grease and oil from the unit. Accumulated grease and oil can cause overheating and subsequent generator damage and may present a potential fire hazard. Do not leave in the generator case or control box anything which does not belong to the assembly. The tightness of lids and covers as well as of all screw joints must be inspected on a regular basis, for instance at least once a week or more frequently, if a generator is exposed to a noticeable vibration. Having carried a maintenance or cleaning, **NEVER LEAVE ALTERNATOR'S COVER AND/OR ELECTRIC BOX' COVER UNREPLACED AND UNSECURED !** Condition of any seal/gasket must be inspected and defective parts replaced.

## GROUNDING DYNASET EQUIPMENT



Dynaset equipment has to ground in cases when rubber cushions or plastic pads are installed to the framework. Grounding is also desirable in cases suspected insufficient grounding between working machine and Dynaset equipment.

Use ground wire size at least phase wire cross section area.

Ground wire cross section area (S / mm<sup>2</sup>)  $\geq$  Phase wire cross section area (S / mm<sup>2</sup>).

Depending on operation conditions Dynaset recommend use larger ground wire size than phase wire cross section area.

Pay attention in Dynaset HG, HWG, HMG and CMG models used with rubber cushions or plastic pads enough grounding does not achieve via hydraulic connections. Enough grounding achieves with external galvanic wire. In fixed installation without rubber cushions or plastic pads we recommend grounding, but it's not mandatory.

Grounding wire length 500mm  $\pm$  100mm, see picture.

Pay attention in working machine, e.g. a floating electric circuit forms by Dynaset generator installation and grounding to machine framework.

This circuit is not in galvanic connection to real earth potential instead they have their own potential in machine's framework.



## HYDRAULIC FLUIDS

Wide range of standard hydraulic fluids can be used with the DYNASET hydraulic equipment.

Depending on the operating temperature, following mineral hydraulic oils are recommended:

|           |                                              |
|-----------|----------------------------------------------|
| ISO VG 32 | for oil's operation temperature up to 70 °C; |
| ISO VG 46 | for oil's operation temperature up to 80 °C; |
| ISO VG 68 | for oil's operation temperature up to 90 °C. |

Synthetic and bio-oils can be used as well if their viscosity characteristics and lubricating efficiency are corresponding to above mineral oils. Automatic transmission fluids and even engine oils can be used, provided that they are allowed to be used in hydraulic system of your carrier machine.

To use special hydraulic fluids a with DYNASET equipment, please be kindly requested to contact nearest DYNASET representative for an advice.

## PRODUCT DISPOSAL

Conform to waste legislation, regulations and recommendations in waste disposal and waste recycling issued by local authorities.

### 1. Precondition:

- Product is permanently useless or beyond repair.
- Before transportation get off all used agent (oil, cooling liquid) and dirty filters.

Items requiring special handling can normally be done by authorized waste management facility if not:

- Separating the base materials, iron, copper, steel, electronics, removing paint, polyester resin, and insulation tape and/or plastics residues from all components.
- This 'waste material' can now be recycled.

### 2. Deliver the recycling and waste material to waste management facility.

Note! Customer can send the DYNASET equipment for reuse or recycling to the Dynaset Oy or to other location determinate by Dynaset representative.

- Customer pays shipping cost.
- Equipment must be adequately packed for shipment.
- Shipment documents must contain purchaser's name, contact information, equipment type and serial number.



**TROUBLESHOOTING**

|                                                                                        |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.<br/>LOW OUTPUT<br/>VOLTAGE AT NO<br/>LOAD</b>                                    | <p>1.1<br/>Too low rotation speed of generator.</p> <p>1.2<br/>Excitation rectifier's failure.</p> <p>1.3<br/>Poor contact in electric system.</p> <p>1.4<br/>Voltage regulator's failure.</p> <p>1.5<br/>Winding failure.</p> | <p>1.1.1<br/>Check first the output frequency. If the frequency is out of range, hydraulic system failure is concerned.</p> <p>1.1.2<br/>Check whether the hydraulic fluid flow and pressure are sufficient. Adjust when necessary.</p> <p>1.1.3<br/>Check the hydraulic motor for possible leakage. Replace motor if necessary.</p> <p>1.2<br/>Trace the failure and replace the rectifier.</p> <p>1.3<br/>Check all internal contacts and wiring of the generator.<br/>Check and clean brushes and slip ring (HG6,5-HG20).</p> <p>1.4.1<br/>Replace the capacitor (HG3,5—HG5,0).</p> <p>1.4.2<br/>Check and adjust the air gap of the compound regulator. Replace if broken (HG6,5 - HG15).</p> <p>1.4.3<br/>Check and adjust or replace the electronic regulator (HG30).</p> <p>1.5<br/>Verify the winding resistance with parameters shown in technical specification and replace if damaged.<br/>ATTN !<br/>Prior to measuring an insulation resistance, set overload protection to OFF-position !</p> |
| <b>2.<br/>LOW OUTPUT<br/>VOLTAGE AT LOAD,<br/>WHILE NO-LOAD<br/>VOLTAGE IS CORRECT</b> | <p>2.1<br/>The generator is being overloaded.</p> <p>2.2<br/>Too low rotation speed of generator.</p> <p>2.3<br/>Voltage regulator's failure.</p>                                                                              | <p>2.1<br/>Reduce the load and check the current I (A) to ensure that the proper load is being applied.</p> <p>2.2.1<br/>Hydraulic pressure insufficient.</p> <p>2.2.2<br/>Hydraulic system failure to be traced and cleared.</p> <p>2.2.3<br/>Hydraulic motor worn out. Replace hydraulic motor.</p> <p>2.3.1<br/>Replace the capacitor (HG3,5 - HG 5,0).</p> <p>2.3.2<br/>Check and adjust or replace the compound regulator (HG6,5 - HG20).</p> <p>2.3.3<br/>Check and adjust or replace the electronic regulator (HG30).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**... TROUBLESHOOTING**

|                                        |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3.</b><br><b>EXCITATION FAILURE</b> | <p>3.1<br/>Excitation rectifier's failure.</p> <p>3.2<br/>Voltage regulator's failure.</p> <p>3.3<br/>Winding failure.</p> <p>3.4<br/>Poor contact in electric system.</p> <p>3.5<br/>Insufficient residual magnetism.</p> | <p>3.1<br/>Trace the failure and replace the rectifier.</p> <p>3.2.1<br/>Replace the capacitor (HG3,5 - HG5,0).</p> <p>3.2.2<br/>Check and adjust or replace the compound regulator (HG6,5 - HG20).</p> <p>3.2.3<br/>Check and adjust or replace the electronic regulator (HG30).</p> <p>3.3<br/>Verify the winding resistance with parameters shown in technical specification and replace if damaged.</p> <p>3.4<br/>Check all internal contacts and wiring of the generator. Check and clean brushes and slip ring (HG6,5 - HG20).</p> <p>3.5<br/>Use external 12 V DC battery for 1 - 2 sec. to magnetise the rotor.</p> |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                |                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4.</b><br><b>OUTPUT VOLTAGE INSTABILITY</b> | <p>4.1<br/>Instable rotation speed of generator.</p> <p>4.2<br/>Electronic voltage regulator's failure (HG30 - HG70).</p> <p>4.3<br/>Poor contact in electric system.</p> | <p>4.1.1<br/>Check generator's hydraulics, including automatic frequency control valve. Make an adjustment if necessary.</p> <p>4.1.2<br/>Check whether the hydraulic fluid flow and pressure are excessive. Adjust when necessary.</p> <p>4.1.3<br/>Check the hydraulic motor for possible leakage. Replace motor if necessary.</p> <p>4.2<br/>Adjust stability of the regulator. Replace if broken.</p> <p>4.3<br/>Check all internal contacts and wiring of the generator. Check and clean brushes and slip ring (HG6,5 - HG20).</p> |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**... TROUBLESHOOTING**

|                                        |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                         |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5.<br/>ABNORMAL NOISE<br/>LEVEL</b> | <p>5.1<br/>Bearing failure.</p> <p>5.2<br/>Generator is being overloaded.</p> <p>5.3<br/>Short circuit in powered unit.</p> <p>5.4<br/>Foreign items in generator's casing.</p> <p>5.5<br/>Extremely fluctuating load.</p> | <p>5.1<br/>Replace broken bearing.</p> <p>5.2<br/>Reduce the load to the proper level.</p> <p>5.3<br/>Check powered unit. Rectify a defect.</p> <p>5.4<br/>Stop generator and hydraulic system. Remove foreign item from the unit.</p> <p>5.5<br/>Can it be fixed ?</p> |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                            |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                               |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6.<br/>OIL LEAKAGES</b> | <p>6.1<br/>Failure of axial sealing of generator's hydraulic motor. External indication - hydraulic oil outflow from ventilation grids.</p> <p>6.2<br/>Oil leakage from hydraulic motor.</p> | <p>6.1<br/>Axial sealing of hydraulic motor broken by reason of EXCESSIVE PRESSURE IN RETURN LINE. Rebuild the return line. Maximum allowed pressure in return line is 5 bar. Replace axial sealing of generator's motor.</p> <p>6.2<br/>Hydraulic motor worn out and should be replaced.</p> |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## PARAMETERS

| HG PARAMETERS                              |             | HG 3,5 kVA<br>115V-17-V                                                                                            | HG 3,5 kVA<br>230V-17-V | HG 5,0 kVA<br>230V-24-V | HG 6,5 kVA<br>400V-33-V | HG 10 kVA<br>400V-48-V | HG 12 kVA<br>400V-57-V | HG15 kVA<br>400V-66-V | HG 20 kVA<br>400V-92-V | HG 20 kVA<br>400V-60-V |
|--------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|
| <b>OUTPUT PARAMETERS at minimum flow</b>   |             |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| Voltage 1                                  | V +/- %     | 115 +/- 6%                                                                                                         | 230 +/- 6%              | 230 +/- 6%              | 400 +/- 5%              | 400 +/- 5%             | 400 +/- 5%             | 400 +/- 5%            | 400 +/- 5%             | 400 +/- 5%             |
| Voltage 2                                  | V +/- %     | 3,5                                                                                                                | 3,5                     | 5,0                     | 230 +/- 5%              | 230 +/- 5%             | 230 +/- 5%             | 230 +/- 5%            | 230 +/- 5%             | 230 +/- 5%             |
| Total power                                | kVA         | 3,5                                                                                                                | 3,5                     | 5,0                     | 6,5                     | 10                     | 12                     | 15                    | 20                     | 20                     |
| Power output                               | kW/cos fi   | 3,5 /1,0                                                                                                           | 3,5 /1,0                | 5,0 /1,0                | 5,2 /0,8                | 8,0 /0,8               | 9,6 /0,8               | 12,0 /0,8             | 16,0 /0,8              | 16,0 /0,8              |
| Frequency                                  | Hz +/- %    | 50 +/- 5%                                                                                                          | 50 +/- 5%               | 50 +/- 5%               | 50 +/- 5%               | 50 +/- 5%              | 50 +/- 5%              | 50 +/- 5%             | 50 +/- 5%              | 50 +/- 5%              |
| Main fuse                                  | recommended | 2 x 16A                                                                                                            | 1 x 16A                 | 1 x 20A                 | 3 x 16A                 | 3 x 16A                | 3 x 16A                | 3 x 25A               | 3 x 32A                | 3 x 32A                |
| <b>FREQUENCY CONTROL</b>                   |             | AUTOMATIC FREQUENCY CONTROL VALVE                                                                                  |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>VOLTAGE CONTROL</b>                     |             | C/I-REGULATOR                                                                                                      |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>SERVICE SOCKETS</b>                     |             | COMPOUND-REGULATOR                                                                                                 |                         |                         |                         |                        |                        |                       |                        |                        |
| 110V                                       | pcs         | 2                                                                                                                  | 2                       | 2                       | 2                       | 2                      | 2                      | 2                     | 2                      | 2                      |
| 230V                                       | pcs         |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| 400V                                       | pcs         |                                                                                                                    |                         |                         | 1                       | 1                      | 1                      | 1                     | 1                      | 1                      |
| 400V                                       | pcs         |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>HYDRAULIC PORTS</b>                     |             |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| Pressure line                              | P           | BSP 1/2"                                                                                                           | BSP 1/2"                | BSP 1/2"                | BSP 1/2"                | BSP 1/2"               | BSP 1/2"               | BSP 3/4"              | BSP 3/4"               | BSP 3/4"               |
| Return line                                | T           | BSP 1/2"                                                                                                           | BSP 1/2"                | BSP 1/2"                | BSP 1/2"                | BSP 1/2"               | BSP 1/2"               | BSP 1"                | BSP 1"                 | BSP 1"                 |
| Service line                               | S           | BSP 3/8"                                                                                                           | BSP 3/8"                | BSP 3/8"                | BSP 3/8"                | BSP 3/8"               | BSP 3/8"               | NO                    | NO                     | NO                     |
| <b>HYDRAULIC REQUIREMENTS</b>              |             |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| Minimum flow                               | l/min       | 20                                                                                                                 | 20                      | 27                      | 36                      | 51                     | 61                     | 70                    | 97                     | 64                     |
| Maximum flow                               | l/min       | 30                                                                                                                 | 30                      | 32                      | 50                      | 70                     | 70                     | 95                    | 115                    | 90                     |
| Pressure at nominal power output, bar      |             | 180                                                                                                                | 180                     | 160                     | 190                     | 190                    | 180                    | 180                   | 180                    | 280                    |
| Maximum pressure                           | bar         | 210                                                                                                                | 210                     | 210                     | 210                     | 210                    | 210                    | 210                   | 210                    | 420                    |
| Idle run pressure                          | bar         | 50                                                                                                                 | 50                      | 50                      | 50                      | 50                     | 40                     | 40                    | 40                     | 30                     |
| <b>HYDRAULIC FLUID REQUIREMENTS</b>        |             | 10 - 200 / optimum 25 - 35                                                                                         |                         |                         |                         |                        |                        |                       |                        |                        |
| Viscosity                                  | cSt         | max. 70 **                                                                                                         |                         |                         |                         |                        |                        |                       |                        |                        |
| Temperature                                | °C          | min. 25                                                                                                            |                         |                         |                         |                        |                        |                       |                        |                        |
| Filter ratio, recommendation               | um          |                                                                                                                    |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>ENCLOSURE PROTECTION</b>                |             | STANDARD                                                                                                           |                         |                         |                         |                        |                        |                       |                        |                        |
|                                            |             | OPTIONAL                                                                                                           |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>EARTH LEAKAGE RELAY</b>                 |             | 1~ / 30 mA   1~ / 30 mA   1~ / 30 mA   3~ / 30 mA   3~ / 30 mA   3~ / 30 mA   3~ / 30 mA   3~ / 30 mA   3~ / 30 mA |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>POWER CABLE OUTLET</b>                  |             | OPTIONAL                                                                                                           |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>SEPARATE ELECTRIC BOARD</b>             |             | OPTIONAL                                                                                                           |                         |                         |                         |                        |                        |                       |                        |                        |
| <b>OIL COOLER, recom. (C) kW / minimum</b> |             | 2                                                                                                                  | 2                       | 3                       | 3                       | 4                      | 5                      | 5                     | 8                      | 7                      |
| <b>WEIGHT</b>                              |             | 26                                                                                                                 | 26                      | 30                      | 42                      | 57                     | 66                     | 98                    | 120                    | 120                    |

## PARAMETERS

| PARAMETERS                                 |             | HG 30 kVA<br>400V - 129           | HG 30kVA<br>400V - 90 | HG 40 kVA<br>400V - 165 | HG 40 kVA<br>400V-90 | HG 50 kVA<br>400V-210 | HG 50 kVA<br>400V-120 | HG 60 kVA<br>400V-120 | HG 70 kVA<br>400V- 165 |  |
|--------------------------------------------|-------------|-----------------------------------|-----------------------|-------------------------|----------------------|-----------------------|-----------------------|-----------------------|------------------------|--|
| <b>OUTPUT PARAMETERS at minimum flow</b>   |             |                                   |                       |                         |                      |                       |                       |                       |                        |  |
| Voltage 1                                  | V + / - %   | 400 + / - 3%                      | 400 + / - 3%          | 400 + / - 3%            | 400 + / - 3%         | 400 + / - 3%          | 400 + / - 3%          | 400 + / - 3%          | 400 + / - 3%           |  |
| Voltage 2                                  | V + / - %   | 230 + / - 3%                      | 230 + / - 3%          | 230 + / - 3%            | 230 + / - 3%         | 230 + / - 3%          | 230 + / - 3%          | 230 + / - 3%          | 230 + / - 3%           |  |
| Total power                                | kVA         | 30                                | 30                    | 40                      | 40                   | 50                    | 50                    | 60                    | 70                     |  |
| Power output                               | kW/cos fi   | 24 / 0,8                          | 24 / 0,8              | 32 / 1,0                | 32 / 0,8             | 40 / 0,8              | 40 / 0,8              | 48 / 0,8              | 56 / 0,8               |  |
| Frequency                                  | Hz + / - %  | 50 + / - 5%                       | 50 + / - 5%           | 50 + / - 5%             | 50 + / - 5%          | 50 + / - 5%           | 50 + / - 5%           | 50 + / - 5%           | 50 + / - 5%            |  |
| Main fuse                                  | recommended | 3 x 40A                           | 3 x 40A               | 3 x 63A                 | 3 x 63A              | 3 x 63A               | 3 x 63A               | 3 x 80A               | 3 x 100A               |  |
| <b>FREQUENCY CONTROL</b>                   |             | AUTOMATIC FREQUENCY CONTROL VALVE |                       |                         |                      |                       |                       |                       |                        |  |
| <b>VOLTAGE CONTROL</b>                     |             | ELECTRONIC REGULATOR              |                       |                         |                      |                       |                       |                       |                        |  |
| <b>SERVICE SOCKETS</b>                     |             | OPTIONAL                          |                       |                         |                      |                       |                       |                       |                        |  |
| <b>HYDRAULIC PORTS</b>                     |             |                                   |                       |                         |                      |                       |                       |                       |                        |  |
| Pressure line                              | P           | BSP 3/4"                          | BSP 3/4"              | BSP 3/4"                | BSP 3/4"             | BSP 1"                | BSP 1"                | BSP 1"                | BSP 1"                 |  |
| Return line                                | T           | BSP 1"                            | BSP 3/4"              | BSP 1 1/4"              | BSP 3/4"             | BSP1 1/2"             | BSP 1 1/2"            | BSP 1 1/2"            | BSP 1 1/2"             |  |
| Service line                               | S           | NO                                | NO                    | NO                      | NO                   | NO                    | NO                    | NO                    | NO                     |  |
| <b>HYDRAULIC REQUIREMENTS</b>              |             |                                   |                       |                         |                      |                       |                       |                       |                        |  |
| Minimum flow                               | l/min       | 135                               | 95                    | 172                     | 95                   | 215                   | 126                   | 126                   | 172                    |  |
| Maximum flow                               | l/min       | 150                               | 110                   | 190                     | 110                  | 230                   | 140                   | 140                   | 190                    |  |
| Pressure at nominal power output, bar      |             | 190                               | 250                   | 190                     | 330                  | 190                   | 350                   | 350                   | 350                    |  |
| Maximum pressure                           | bar         | 250                               | 420                   | 250                     | 420                  | 250                   | 420                   | 420                   | 420                    |  |
| Idle run pressure                          | bar         | 30                                | 30                    | 30                      | 30                   | 30                    | 30                    | 30                    | 30                     |  |
| <b>HYDRAULIC FLUID REQUIREMENTS</b>        |             | 10 - 200 / optimum 25 - 35        |                       |                         |                      |                       |                       |                       |                        |  |
| Viscosity                                  | cSt         | max. 70 **                        |                       |                         |                      |                       |                       |                       |                        |  |
| Temperature                                | °C          | min. 25                           |                       |                         |                      |                       |                       |                       |                        |  |
| Filter ratio, recommendation               | um          |                                   |                       |                         |                      |                       |                       |                       |                        |  |
| <b>ENCLOSURE PROTECTION</b>                |             | STANDARD                          |                       |                         |                      |                       |                       |                       |                        |  |
| <b>EARTH LEAKAGE RELAY</b>                 |             | OPTIONAL                          |                       |                         |                      |                       |                       |                       |                        |  |
| <b>POWER CABLE OUTLET</b>                  |             | STANDARD                          |                       |                         |                      |                       |                       |                       |                        |  |
| <b>SEPARATE ELECTRIC BOARD</b>             |             | OPTIONAL                          |                       |                         |                      |                       |                       |                       |                        |  |
| <b>OIL COOLER, recom. (C) kW / minimum</b> |             | 5                                 | 5                     | 5                       | 5                    | 5                     | 5                     | 5                     | 5                      |  |
| <b>WEIGHT</b>                              |             | 178                               | 178                   | 198                     | 198                  | 249                   | 249                   | 283                   | 283                    |  |

**MANUFACTURER'S LIMITED WARRANTY****1. Warranty coverage**

All hydraulic accessories manufactured by DYNASET OY are subject to the terms and conditions of this limited warranty. Products are warranted to the original purchaser to be free from defects in materials or workmanship. Exclusions from warranty are explained in item 8.

**2. Beginning of warranty period**

Warranty period begins from the delivery date of the product. Delivery is considered to be done on the date when installation has been accomplished or purchaser has taken the product in use. Product is considered as taken in use at the date when DYNASET OY has delivered the product to purchaser, unless separately agreed otherwise by written agreement.

**3. Warranty period**

Warranty period is twenty four (24) months based on maximum of 2000 hours usage during this time period. In cases where the system is provided complete with certain special components (e.g. drive unit), those components are considered as a subject to their manufacturer's warranty.

**4. Warranty procedures**

Immediately upon identifying a problem which purchaser believes to be a failure subject to the product's limited warranty, purchaser must contact primary to the seller of the product. Contact must be made as soon as possible, latest thirty (30) days after the problem was identified. Seller and/or manufacturer technical staff determines the nature of the problem primarily by phone or e-mail. Purchaser commits to give necessary information and to perform routine diagnostic procedures in order to determine the nature of the problem and necessary procedures.

**5. Warranty repairs**

If the product is found to be defective during the warranty period, DYNASET OY will, at its option, either repair the product, author it to be repaired at its authorized workshop or exchange the defective product. If the product must be repaired elsewhere than premises of DYNASET OY or authorized workshop, all costs excluded from this warranty (traveling and waiting hours, daily allowance, traveling expenses and uninstallation/reinstallation costs) will be charged from the purchaser.

If the problem is not covered by this limited warranty, DYNASET OY has the right to charge purchaser of troubleshooting and repairing.

**6. Delivery terms of warranty repair**

If the product is found possible to be defective under this limited warranty and it needs to be repaired, DYNASET OY gives Warranty Return Number (WRN). Items being returned must be shipped, at the purchaser's cost, adequately packed for shipment, to the DYNASET OY or to other location authored by DYNASET OY.

Shipment documents must contain:

- Purchaser's name and contact information
- Receipt of original purchase
- WRN code
- Problem description

**7. Warranty of repaired product**

Warranty period of the product repaired under this limited warranty continues to the end of original warranty period.

**8. Exclusions from warranty**

This warranty shall not apply to:

- a. Failures due to normal wear and tear, improper installation, misuse, abuse, negligence, purchaser selection of improper product to intended use, accident, improper filtration of hydraulic oil or intake water or lack of maintenance
- b. Cost of maintenance, adjustments, installation or startup
- c. Coating, hydraulic oil, quick couplings and interconnection hoses (internal or external to system assemblies)
- d. Products altered or modified in a manner not authorized by DYNASET OY in writing
- e. Products which have been repaired during warranty period by others than DYNASET OY or its authorized workshop
- f. Costs of any other damage or loss, whether direct, indirect, incidental, special or consequential, arising out of the use of, or the inability to use, the product
- g. Telephone or other communications expense
- h. Product that is used in exceptional conditions, considered to cause excessive wear and tear
- i. Faults caused by nature phenomenon's like flood, thunder, etc.

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## EC DECLARATION OF CONFORMITY

We hereby declare that the below-reference product design, structure and manufacturing are in conformity with the provisions in the Council Directive on mutual approximation of laws of the Member States on the safety of machines.

- Machine directive 2006/42/EC
- LVD directive 2006/95/EC
- EMC directive 2004/108/EC.

Applied harmonized standards:

- EN60204-1 Safety of machinery – Electrical equipment of machines.
- EN ISO 4413: EN ISO 4413:2010 Hydraulic fluid power - General rules and safety requirements for systems and their components.

If device has changed by someone other than at the hands of manufacturer or his permission, this declaration is not valid.

**PRODUCT:** DYNASET HYDRAULIC GENERATOR FAMILY

**ALL TYPE NUMBERS:** HGxxxx-xxxxxxxx-xxxx-xxxx

YLÖJÄRVI 31.08.2012

**DYNASET Oy**



Timo Nieminen  
R&D Manager

## NOTES

[illegible]